AEROGUARD AI

Next-Gen Zero-Trust Endpoint Security & Behavioral Analytics Engine

DOCUMENT VERSION: 1.0 (V1 BLUEPRINT) | **TARGET VALUATION (YEAR 1):** ₹1 CRORE – ₹3 CRORE | **SECURITY LEVEL:** ENTERPRISE GRADE

Executive Summary: AeroGuard AI is a lightweight, zero-cloud endpoint security system engineered to safeguard corporate assets against insider data theft, phone-camera leaks, and unauthorized station access. Operating at system-level hardware hooks with 0% cloud dependency, it delivers unbreakable compliance and continuous biometric verification without degrading hardware performance.

1. THE PROBLEM (ENTERPRISE SECURITY VULNERABILITIES)

Modern enterprise endpoint defense solutions suffer from critical architecture flaws that expose companies to millions of dollars in compliance fines and intellectual property theft:

- **High CPU & Cloud Dependency:** Legacy tools like CrowdStrike or Symantec hog system resources, slowing down low-to-mid range PCs while rendering protection useless when offline.
- **The "Camera Leak" Exploit:** Employees routinely bypass software copy/paste blocks by taking photos of sensitive screens via personal mobile devices—a loophole unaddressed by traditional firewalls.
- **Static Session Locks:** Once an endpoint is unlocked, any physical intruder can exfiltrate sensitive files before automated OS sleep triggers kick in.

2. THE AEROGUARD SOLUTION (CORE ARCHITECTURE PILLARS)

PILLAR	TECHNICAL MECHANISM	CORE ADVANTAGE
Pillar 1: Keyboard Rhythm Analytics	Tracks Keystroke Dynamics via Dwell Time (T_d) & Flight Time (T_f) with millisecond precision.	100% Offline biometric identity verification. Instantly locks or flags session if rhythm match drops below 85%.
Pillar 2: Dynamic Stealth Watermark	Kernel/Graphics-level overlay displaying dynamic User ID, Local IP, & Date-Time with crypto steganography.	Defeats phone camera leaks. Invisible encrypted micro-tags trace cropped screen photos back to the perpetrator.
Pillar 3: Zero-Knowledge Vault	Sub-system local SQLite database secured via AES-256 encryption using local Hardware Fingerprinting.	Zero master key leakage on cloud servers. Hard disk extraction yields unreadable garbage data.

3. VERSION 1 (V1) SCOPE & TECHNICAL DELIVERABLES

- **Module 1 — Silent Keystroke Listener:** Background listener utilizing system-level hooks to extract key-gap timing matrices without recording actual key text (GDPR/Privacy Compliant).
- **Module 2 — Dynamic Screen Overlay:** Lightweight Python/Win32 GUI layer rendering tamper-proof, semi-transparent watermark metrics over active screen view.

- **Module 3 — Fail-Safe Watchdog & Logging:** Dual-process architecture (`Watchdog Daemon`) providing instant auto-restart upon unexpected main engine crashes, wrapped in isolated `try-except-finally` code blocks.
- **Module 4 — Encrypted Local Logger:** Encrypted SQLite transaction engine storing anomaly alerts locally for automated weekly admin sync.

YEAR 1 VALUATION TARGET

₹1,00,00,000 – ₹3,00,00,000

Based on 1,000 Active Endpoints @ ₹300/User/Month ARR Model

4. 12-MONTH COMMERCIALIZATION ROADMAP

Months 1–3: V1 PROTOTYPE Core engine completion, local testing, zero-cost development, and initial IP filing.

Months 4–6: BETA DEPLOYMENT Onboarding 5–10 local IT firms (approx. 250 endpoints) for field validation. Valuation: ₹50L – ₹75L.

Months 7–12: COMMERCIAL SCALE B2B enterprise sales launch (₹300/PC/mo), Code Signing Certificate integration, & ARR growth. Valuation: ₹1Cr+.

5. TECHNICAL STACK & ZERO-COST ARCHITECTURE

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Language & Logic   : Python 3.x (Core Prototype) -> C++ / Rust (Kernel Hooks Scale)
UI Overlay Engine  : Win32GUI / PyGObject / Tkinter
Database & Cipher  : SQLite3 + Cryptography (AES-256 GCM Mode)
Deployment Target  : Standalone Executable (.exe / .msi Package via PyInstaller/Nuitka)
Development Cost   : ₹0 (100% Open-Source & Local Workstation Setup)
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6. RISK MITIGATION & PRODUCTION FAIL-SAFE

To eliminate client PC freeze or crash risks, AeroGuard AI implements a dual safety layer:

- **Remote Kill-Switch:** Reads a local network `security_policy.json` file every 5 minutes. IT administrators can toggle off features instantly across all 5,000 PCs without requiring a system reboot.
- **Graceful Degradation:** If typing biometric analysis fails, the system isolates the module and defaults to active screen watermarking without bringing down the OS.